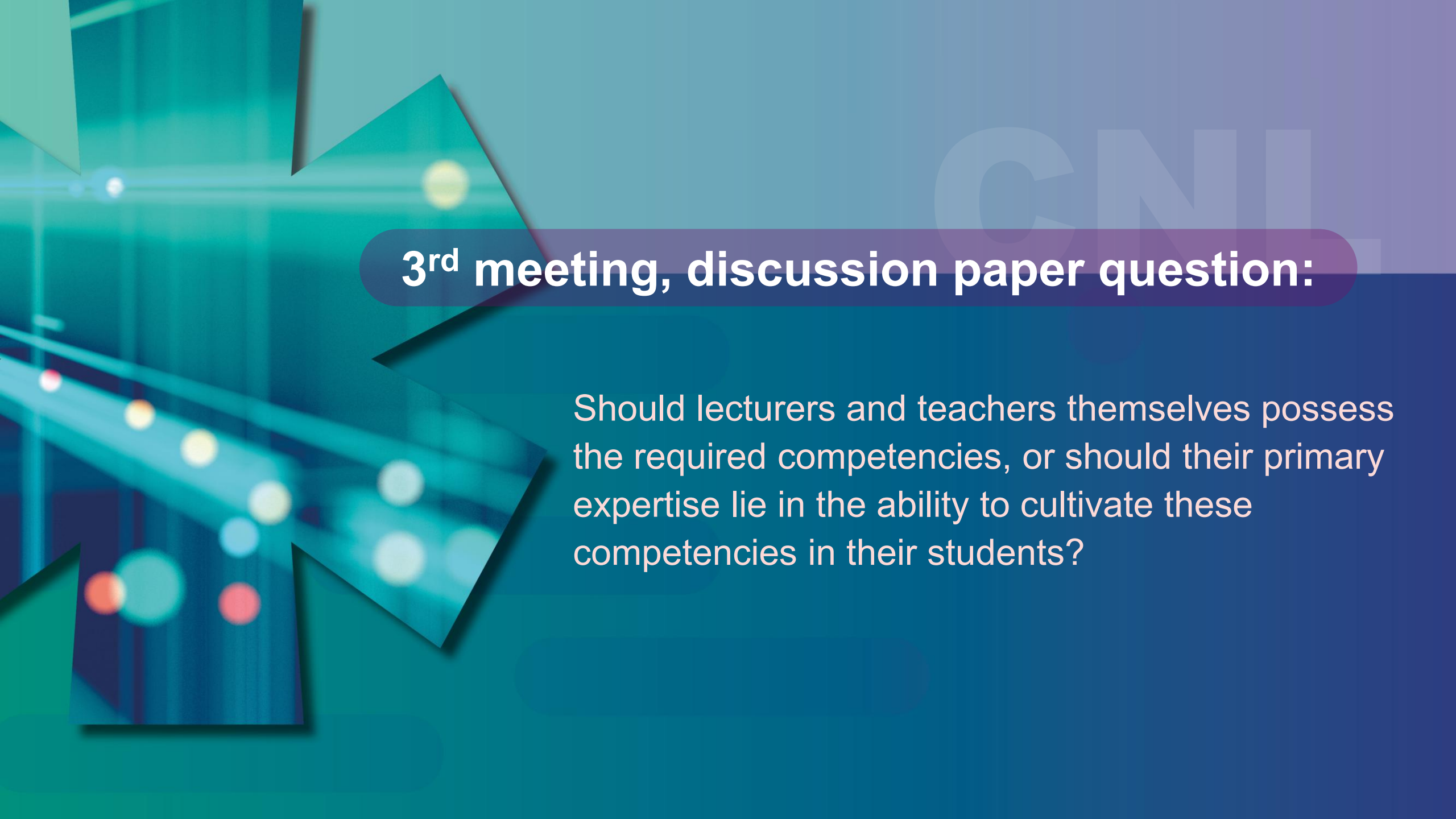


Educators at the times of NEW LEARNING

International STEM Skills Round Table, Phase III, 10.03.26

Dr. Olena Bekh, Senior Human Capital Development Expert -
Team Leader for Innovative Teaching and Learning





3rd meeting, discussion paper question:

Should lecturers and teachers themselves possess the required competencies, or should their primary expertise lie in the ability to cultivate these competencies in their students?

About us: European Training Foundation (ETF)

28 partner countries

The ETF, an agency of the **European Union**, has a focus on **human capital development**.

We support the **reform of education, training, and labour market systems in 28 partner countries**, in the context of the **EU external relations**.

We gather data, develop knowledge and offer **policy advice** using diagnostic and assessment methodologies, policy tools and frameworks.

We are based in **Turin, Italy**



THEMATIC DOMAINS 2026:



Educators: teachers and trainers



Curricula and key competence



Digital transition of education



CNL RESEARCH

ETF Creating New Learning initiative 2025-2026

In 2020-2025 we conducted two rounds of literature review with case studies and cross-case analysis - to identify relevant trends and innovations in teaching and learning.

Initial literature reviews 2020 are on the ETF Open Space:

[Digital and Online Learning \(DOL\)](#)

[Learning environments](#)

[Personalised and differentiated learning](#)

[Curricula and key competences](#)

[Educators and educators' qualifications](#)

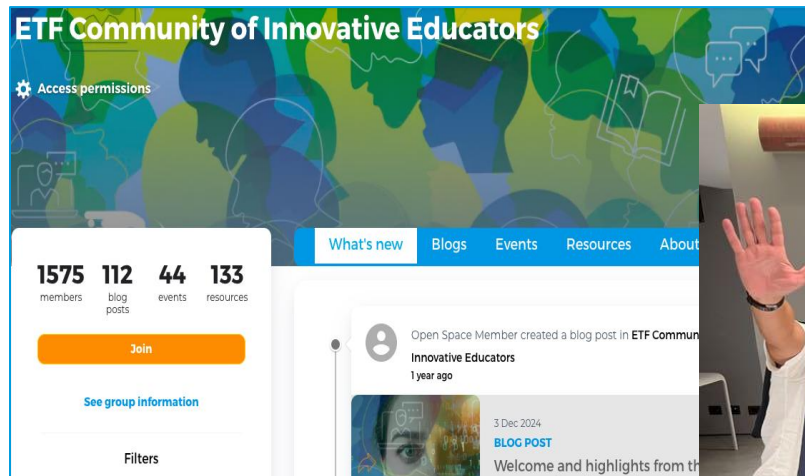
Participatory research was conducted which resulted in 18 case studies from 8 ETF partner countries (including Israel).

A new publication is being prepared for launching in spring 2026.



COMMUNITY OF INNOVATIVE EDUCATORS

The Club “Teaching with AI” is part of the ETF Community of Innovative Educators



Join the community to:

- stay updated on next activities
- be part of one or more Club
- share your innovative teaching practice
- access the knowledge base of post, past events, etc...

<https://openspace.etf.europa.eu/pages/etf-community-innovative-educators>

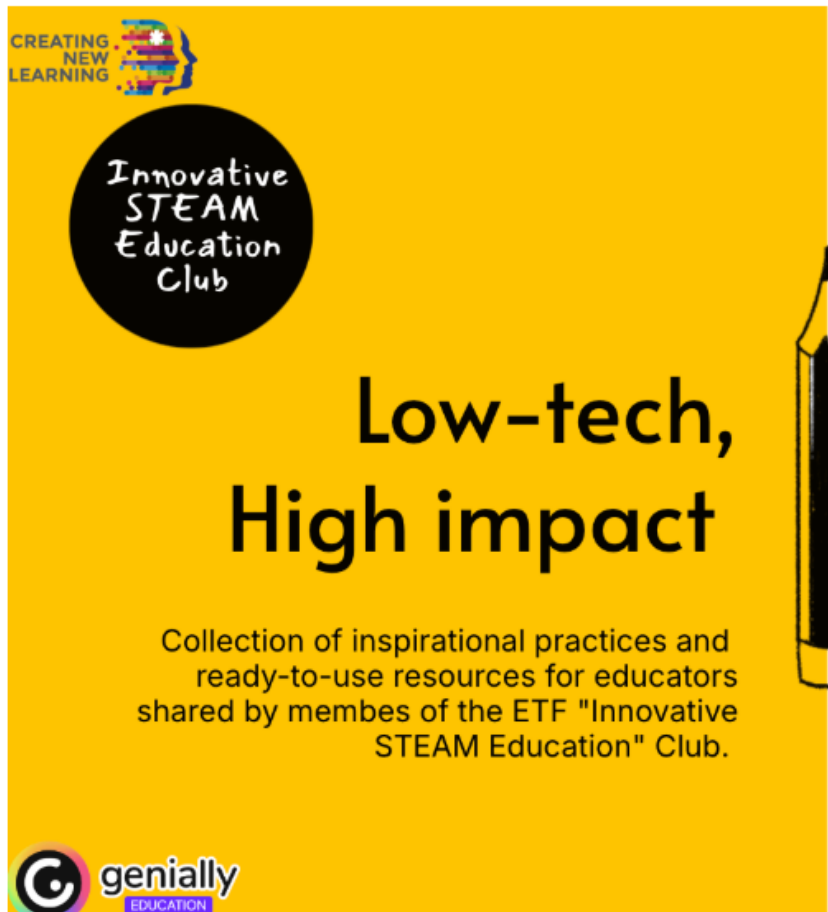
[New Learning Club on Teaching with Artificial Intelligence | Open Space](#)

[TOOL: Unlocking AI in education: A Teacher's Guide to AI Literacy | Gamma](#)

The Club “Teaching with AI” is part of the ETF Community of Innovative Educators

You can explore the **interactive tool** presented during the meeting at this link: [Low-tech, High-impact - STEAM Club output](#) 

The tool is now available also in Georgian language! Discover it [HERE](#) .



How to make a start
with meaningful
STEAM at school?

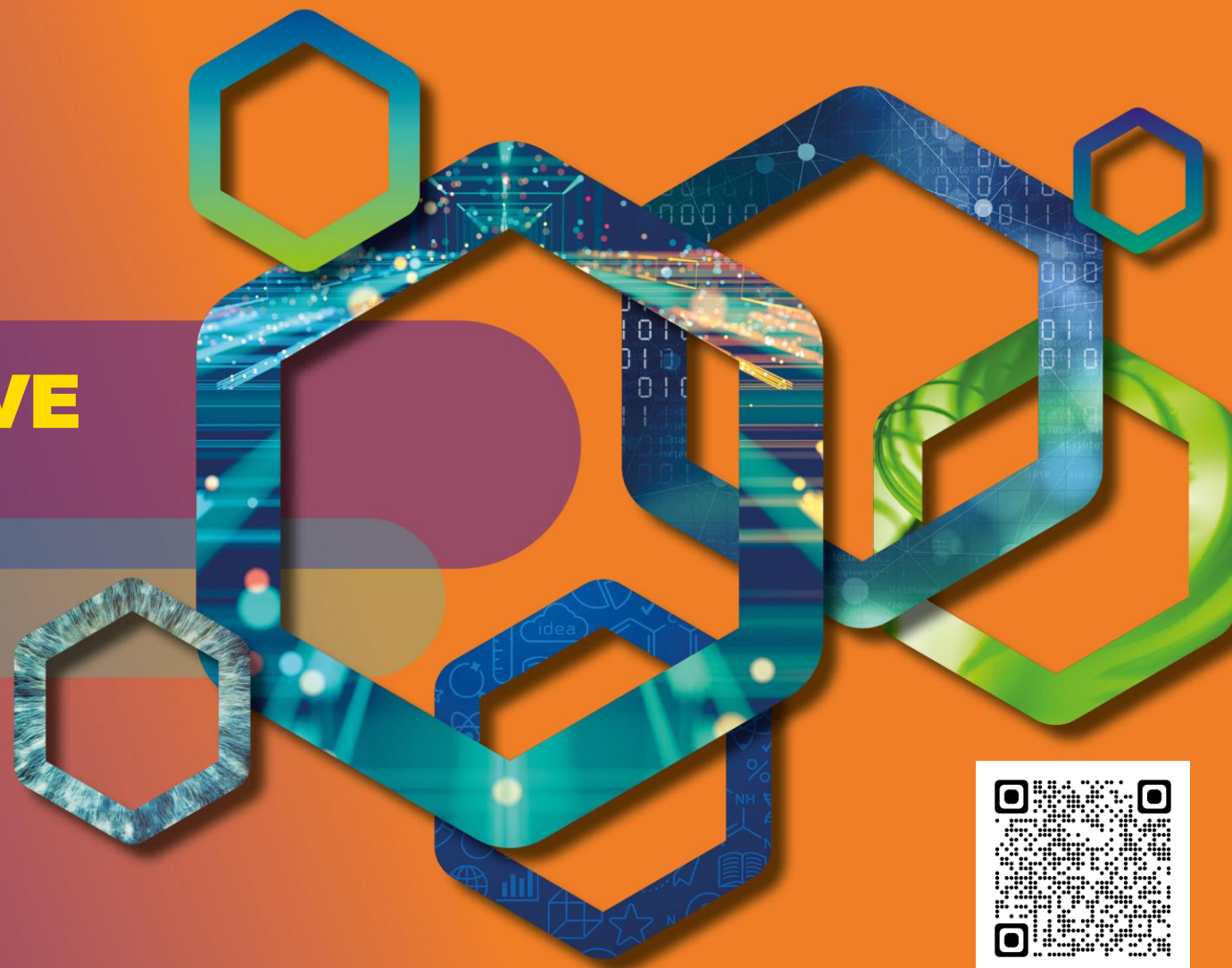
start

[New Learning Club on “Innovative STEAM education” | Open Space](#)

Tool: <https://view.genially.com/67e11174639c71a19a9d82ee>

CREATING NEW LERNING (CNL)

TOOLS FOR INNOVATIVE EDUCATORS



“SELFIE” &
“SELFIE” for Teachers

“READY” Model
for 21° century educators

Guidelines and tools:

- **“Teacher Booster”** videos
- **“Scaffold”** Card Deck
- **Video Pedagogy** Guides

CNL'S TOOLS FOR EDUCATORS

Micro credentials for
Ukraine's Emergency Support

The **ETF's Community**
of Innovative Educators

**Formative
Assessment** Toolkit



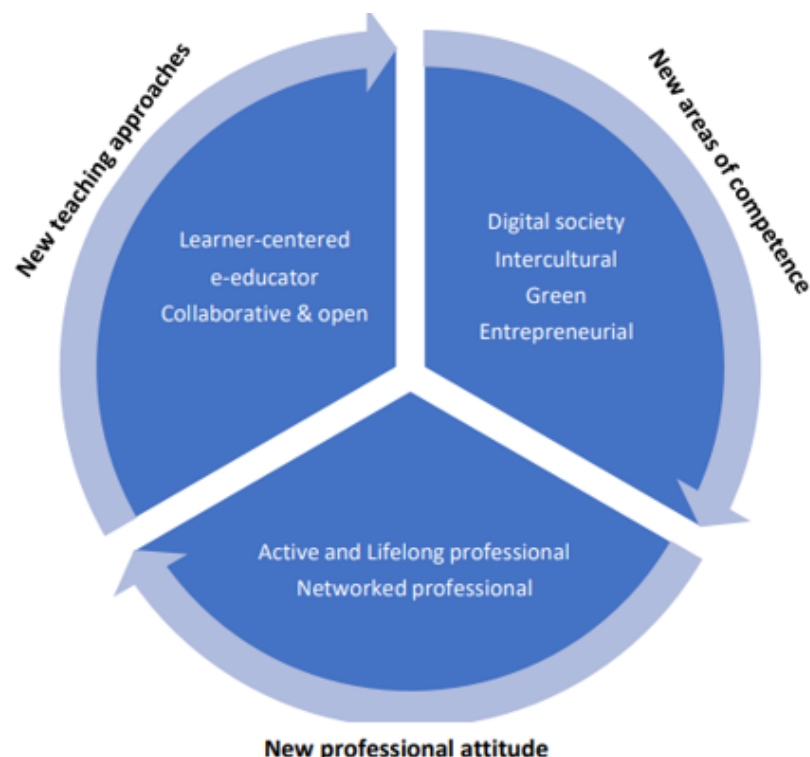
READY (REFERENCE MODEL
FOR Educators' Activities and Development in
THE 21ST-CENTURYY)

- The READY model aims to offer a structured way to identify the **professional practices** and **development needs** of 21st-century educators.

- The role of the contemporary educator is evolving from a "gate-keeper of knowledge" to a "gate-opener to knowledge, skills and attitudes".

** The model was first piloted in Israel and Ukraine*

Emerging teaching approaches and competence areas – where we started in 2020:



[ETF Research report Educators.pdf](#)

Table 2. Impact of drivers of change on VET systems and on VET educators, as emerging from literature review

Trend	Impact on VET systems	Impact on VET educators	Sources
1. Digital transformation, including industry 4.0	The developments in digital technology, including Industry 4.0, Internet of Things (IoT), Artificial Intelligence (AI), smart production and big data, not only affect the skills demands in labour market, but also change the profile of jobs: VET systems need to adapt to cater for these changes.	Educators need to acquire both the capacity to use ICT-enhanced pedagogy and the critical knowledge about digitalisation issues such as industry 4.0, AI and IoT. They can use ICT to monitor and coach learners across different contexts.	Tyler et al. 2017, Avis 2017, Wagiran et al. 2019, Thurlings 2015, Brown et al. 2019, Fullan et al. 2014, Axmann et al. 2015, ETF 2019, Misra 2011, Marope et al. 2015, Downes 2012, Bates 2015, Subrahmanyam 2020, Spöttl et al. 2016
2. New teaching paradigms and approaches, including Competence-Based Education (CBE)	New teaching approaches are increasingly being implemented within VET, and learning environments are consequently widening to include multiple learning venues such as classrooms, workshops and online settings. CBE approaches are having a radical impact on VET system in terms of curricula and teaching methods.	Educators need to be able to apply new teaching approaches, adapting them to their contexts. CBE and personalised learning call for higher integration of theory and practice and for more teamwork among educators.	Brundiers and Wiek 2013, Anane and Okwabi 2013, Gulikers et al. 2018, Zyrianova et al. 2018, ETF 2019, Fullan et al. 2014, Axmann et al. 2015
3. Migrations and changing demography	Migration phenomena increase the diversity of VET students, adding up to aging and changing demography dynamics: VET systems need to be oriented not only to local skills supply, but also to migrating populations, including international workers.	Educators need to acquire intercultural communication and linguistics competences, as well as the capacity to support an increasingly aging VET learners population.	Teräs and Lasonen 2013, Finnish National Board of Education 2009, Nakar 2013, ETF 2019, Van Middlekoop et al. 2017, Axmann et al. 2015, Marope et al. 2015, Tran and Pasura 2018
4. Climate change	The transition towards sustainable economies is placing pressure on VET systems to prepare the labour force for green jobs, and calls therefore for the development of green skills. Also, a green awareness is necessary for all jobs to ensure environmentally friendly workplace practices.	Educators need both to be knowledgeable about green economy developments and to have a sustainable and green attitude in their work.	Leicht et al. 2018, Pavlova 2019, Cotton and Winter 2010, Tilbury 2011, Sevilla and Dutra 2018, Brown et al. 2019, Wesselink and Wals 2011
5. New forms of entrepreneurship	The need for entrepreneurial capacity development across sectors as well as the new forms of entrepreneurship (circular economy, crowdfunding, etc.) are having an impact in the world of work and need to be embedded in VET provision.	Educators need to be able to develop entrepreneurship capacities among learners and need to be aware about new form of entrepreneurship.	Avis 2017, Zhang et al. 2017, Roberts and Owen 2012, Bacigalupo et al. 2016, Van Dam et al. 2010, Neck et al.
6. Increased networking and collaboration	Collaborative ways of working can bring added value to both professional and personal life. Professionalism and expertise are no longer understood as personal properties, but closely tied to communities and networks.	Educators need to become “connected professionals”, capable to work through professional networks and to leverage open communities.	Thurlings 2015, Roberts and Owen 2012, Finnish National Board of Education 2009, Lieberman and Mace 2009, Marope et al. 2015

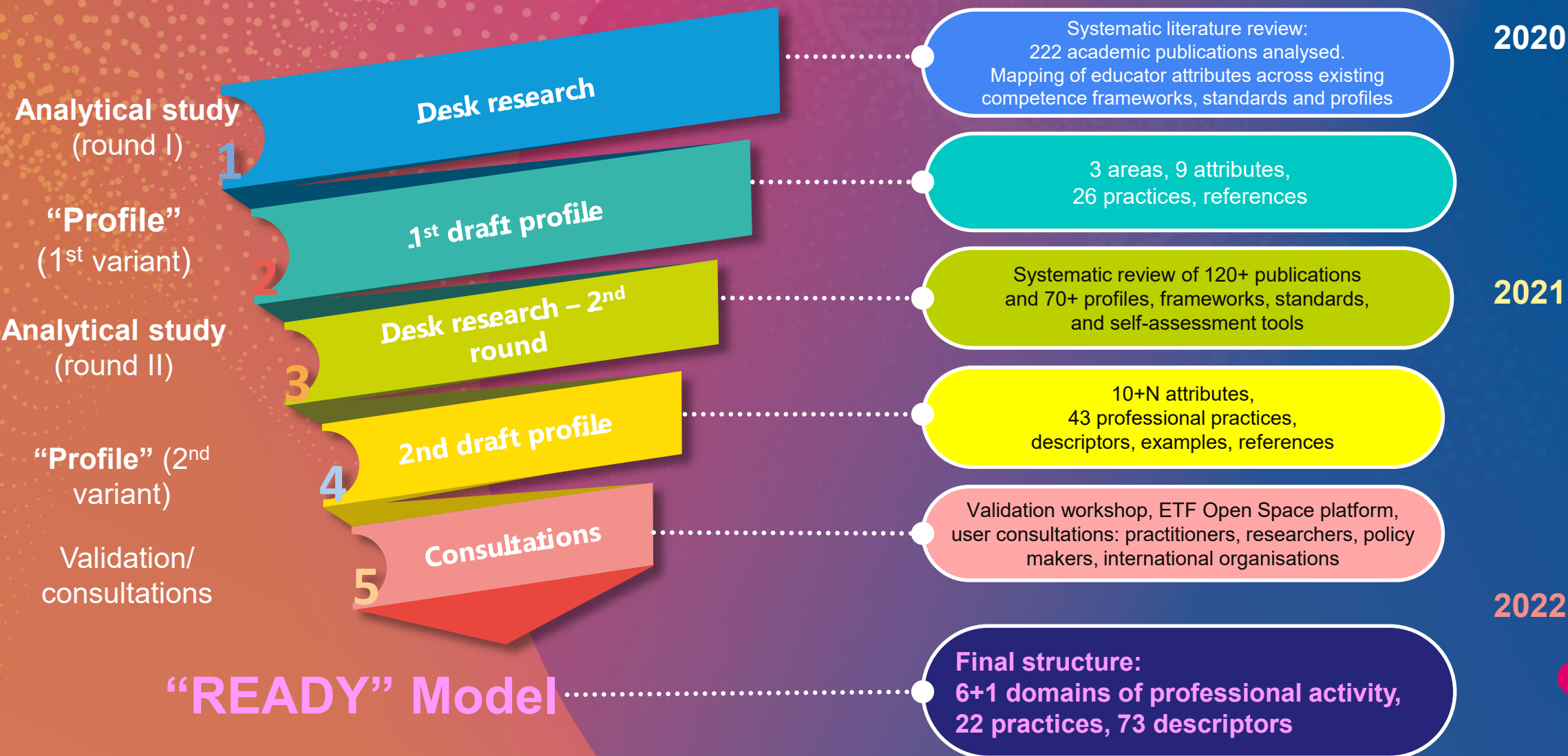
INTERACTIVE “READY” MODEL

120+ publications and 70+ profiles, frameworks, Standards and self-assessment tools

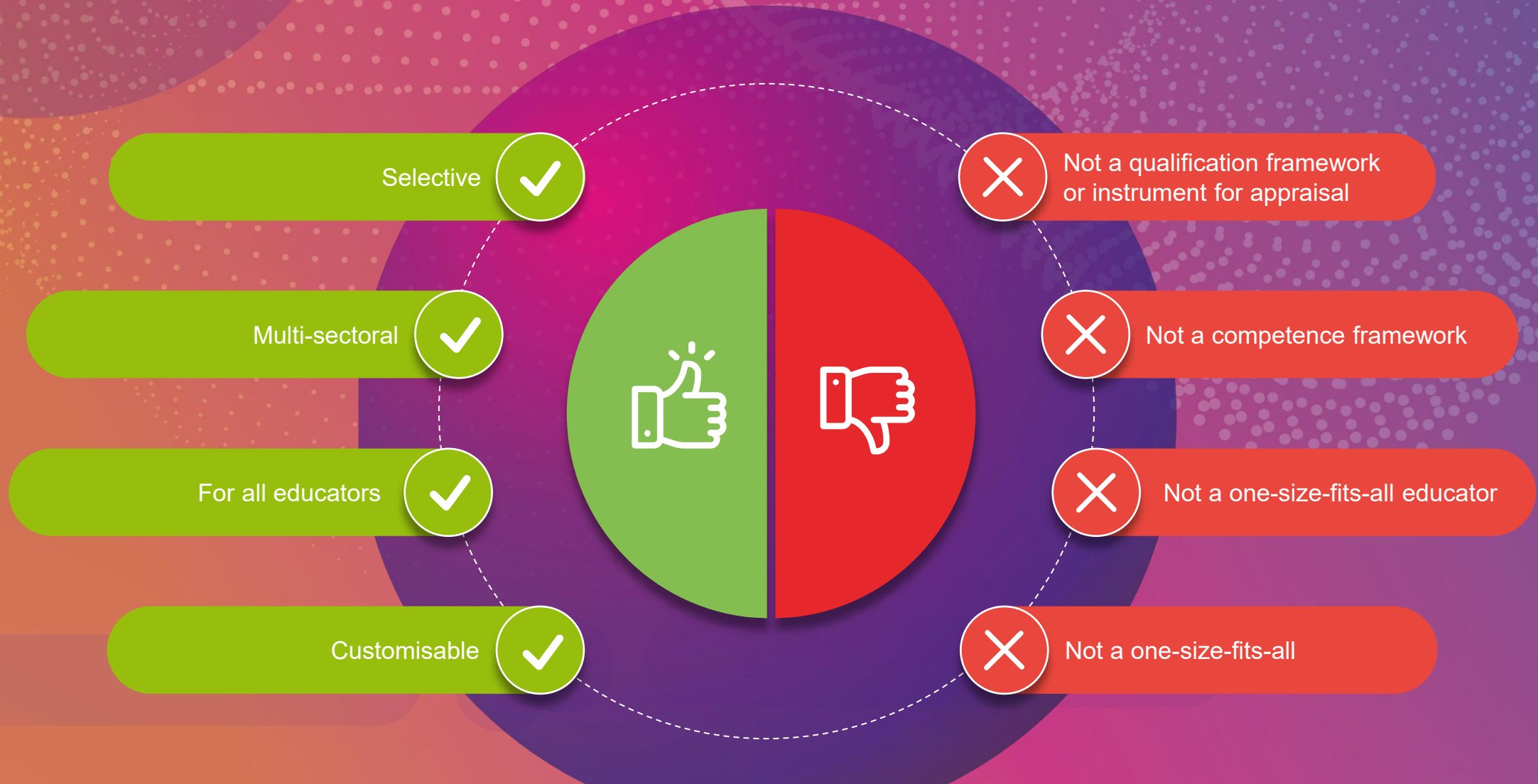


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METHODOLOGY



A REFERENCE MODEL – NOTHING MORE, NOTHING LESS!



• FOR EDUCATORS

- Provide a set of successful professional cases
- Identify strengths and areas (directions) that need improvement
- Identify and plan professional development needs

FOR INSTITUTIONS AND SUBJECTS OF PROFESSIONAL DEVELOPMENT

- Identify and plan professional development needs
- Provide feedback to educators
- Update primary (pedagogical) education programs and job descriptions of educators
- To develop an integrated approach to the qualification of educators
- Review organizational strategies and prioritize required skills
- Improve hiring procedures

FOR POLICYMAKERS AND MANAGERS

- To provide conditions for cooperation between modern educators (common language)
- Create support and professional development systems
- Inspire future teachers to choose a profession
- Provide a national/regional/local point of reference for professional development
- To increase public perception and understanding of the modern teaching profession
- Promote teacher education and continuing professional development programs



READY'S STRUCTURE

A

6+1 domains

They characterise the 21st-century educator

B

22 professional practices

Derived from literature and existing conceptual models

C

73 descriptors

They illustrate how practices can be implemented in real settings

DOMAINS

PRACTICES

DESCRIPTIONS

See our article in the “Journal of e-Learning and Knowledge Society”:
https://www.je-lks.org/ojs/index.php/Je-LKS_EN/article/view/1135980

-  READY model
-  What is it about
-  Why & how it has been created
-  Who is it for
-  How to adapt & use it
-  Resources
-  Contact & Contribute

 **Learner-driven** 

[+ INFO](#)

-  **Foster meaningful learning** 
-  **Facilitate peer- and team-learning** 
-  **Apply a variety of assessment methods** 
-  **Coach learners across learning environments** 
-  **Develop personalised learning experiences** 

	Adaptability & Initiative	
	Sustainability & Inclusion	
	Collaboration & Engagement	
	Lifelong learning & Reflection	
	Digital technologies	
	[Context specific]	

-  READY model
-  What is it about
-  Why & how it has been created
-  Who is it for
-  How to adapt & use it
-  Resources
-  Contact & Contribute

	Learner-driven	+
	Adaptability & Initiative	-

+ INFO

-  Foster learners' entrepreneurship
-  Bridge the worlds of education and work
-  Demonstrate resilience and adaptability

	Sustainability & Inclusion	+
	Collaboration & Engagement	+
	Lifelong learning & Reflection	+
	Digital technologies	+
	[Context specific]	+

-  READY model
-  What is it about
-  Why & how it has been created
-  Who is it for
-  How to adapt & use it
-  Resources
-  Contact & Contribute

	Learner-driven	+
	Adaptability & Initiative	+
	Sustainability & Inclusion	-

+ INFO

- Create inclusive learning environments 
- Be attentive to personal well-being and that of others 
- Manage one's own and others' emotions 
- Adopt and promote environmental, social and economic sustainability 

	Collaboration & Engagement	+
	Lifelong learning & Reflection	+
	Digital technologies	+
	[Context specific]	+

RETURN

HOME

 READY model

 What is it about

 Why & how it has
been created

 Who is it for

 How to adapt
& use it

 Resources

 Contact &
Contribute



Learner-driven



Adaptability & Initiative



Sustainability & Inclusion



Collaboration & Engagement



+ INFO

○ **Collaborate with peers and other stakeholders**



○ **Contribute to professional networks and
communities**



○ **Contribute to organisational development and
improvement processes**



Lifelong learning & Reflection



Digital technologies



[Context specific]



-  READY model
-  What is it about
-  Why & how it has been created
-  Who is it for
-  How to adapt & use it
-  Resources
-  Contact & Contribute

	Learner-driven	+
	Adaptability & Initiative	+
	Sustainability & Inclusion	+
	Collaboration & Engagement	+
	Lifelong learning & Reflection	-

+ INFO

-  Proactively engage in professional development opportunities
-  Reflect on professional practices
-  Apply evidence-based pedagogies

	Digital technologies	+
	[Context specific]	+

-  READY model
-  What is it about
-  Why & how it has been created
-  Who is it for
-  How to adapt & use it
-  Resources
-  Contact & Contribute

	Learner-driven	+
	Adaptability & Initiative	+
	Sustainability & Inclusion	+
	Collaboration & Engagement	+
	Lifelong learning & Reflection	+
	Digital technologies	-
○ Use digital technologies to enhance teaching  ○ Create digital resources and content  ○ Be aware of copyright and online privacy  ○ Understanding the potential role and impact of digital technologies 		
	[Context specific]	+

+ INFO

So, who are modern educators?....

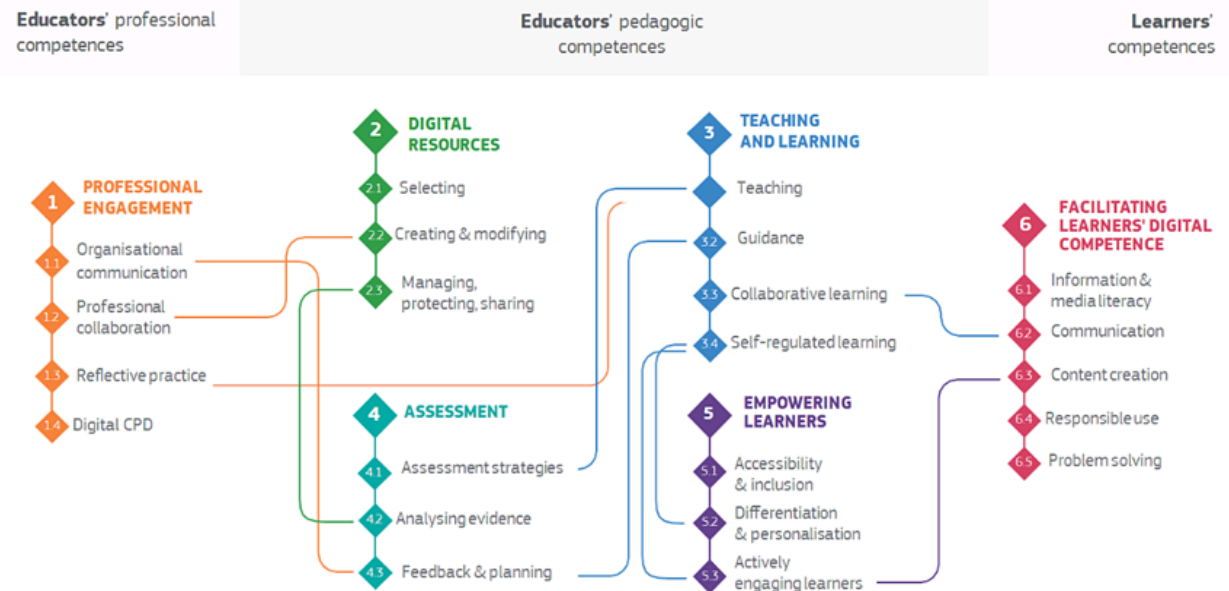
- [ISTE* Standard | 2. Educators:](#)
- 2.1 Learner;
- 2.2. Leader;
- 2.3 Citizen;
- 2.4 Collaborator;
- 2.5 Designer;
- 2.6 Facilitator;
- 2.7 Analyst
- * International Society for Technology in Education

	continuing training		learning)
12	Hattie 2009	Model	Educators
13	ISTE Educator Standards	Standards	Educators
14	Kolb Educator Role Profile	Profile	Educators
15	Green Skills in Vocational Teacher Education – a model of pedagogical competence for a world of sustainable development	Green skills in vocational teacher education	VTE Vocational Teacher Education
16	Pedagogical, Ethical, Attitudinal and Technical dimensions of Digital Competence in Teachers	Model	Teachers

3	DigComp 2.1	Competence framework	Citizens
4	GreenComp		
5	DigComp		
6	Analytical Process-based Method for Teachers' Competence Development		
7	Competence Development		
18	Professional Competence for Teachers		
19	Profile of a County Educator		
20	QESS Professional Competence		
21	San Antonio School District Profile		
22	Teacher competencies and teamwork competencies		
23	Teachers' Professional Competencies	Professional competences	Teachers

	VET Practitioner Capability Framework	(capability)	educators
28	Charles Darwin University VET Educator Capability Framework	Framework (capability)	VET educators
29	Adult Education Teacher Competencies	Competence framework	Adult education

DigCompEdu describes 22 competences organised in six Areas.





Discussion paper question:

Should lecturers and teachers themselves possess the required competencies, or should their primary expertise lie in the ability to cultivate these competencies in their students?

1. The Five Competencies

1. Critical thinking
 2. Teamwork and Interpersonal Communication
 3. Self-Directed Learning
 4. Solving Complex problems
 5. Ethical Judgement and Responsive use of AI
- None of these competences are specific to educators but should be common to 21st century citizens...
 - This list could/should be longer [teachers are citizens]



EU TRANSVERSAL key competences in various contexts

- ✓ Critical thinking
- ✓ Problem solving
- ✓ Teamwork
- ✓ Communication and negotiation
- ✓ Analytical skills
- ✓ Creativity
- ✓ Intercultural skills

2.1 What counts as sufficient mastery?

2.2 Can pedagogical expertise substitute for personal practice?

A. Teachers do not need to be experts...

Vs.

B. Superficial mastery leads to superficial modeling

A. Deep understanding of competency

Vs.

B. Practical experience

- The above are not alternatives... role- and context- specificity...
- Teachers are not super-humans... knowledge base/skills/attitudes
- Teachers as lifelong learners, progression model
- Autonomy & competence /internal QA/teacher roles
- Teach as we preach
- Assessment as learning, self-reflection
- Is learner agency our objective?



2.3 Do all five competences require the same depth of teacher experience?

...Or some competencies are more dependent on lived practices than others?

Yes, and the depth vis-à-vis knowledge/skills/attitudes will vary in each.
Knowledge base..

No. Being a dynamic combination of knowledge, skills and attitudes, they differ by nature and by the share of K/S/A within their learning outcomes. Definition of individual competences (in various frameworks and standards) are theoretical abstracts, outcome of expert compromise...

EU EntreComp framework example

			Level of proficiency		Foundation		Intermediate		Advanced		Expert		
			Progression	Relying on support from others		Building independence		Taking responsibility		Driving transformation, innovation and growth			
					Under direct supervision.	With reduced support from others, some autonomy and together with my peers.	On my own and together with my peers.	Taking and sharing some responsibilities.	With some guidance and together with others.	Taking responsibility for making decisions and working with others.	Taking responsibility for contributing to complex developments in a specific field.	Contributing substantially to the development of a specific field.	
					Discover	Explore	Experiment	Dare	Improve	Reinforce	Expand	Transform	
Area	Competence	Hint	Descriptor	Thread ¹⁰	Level 1	Level 2	Level 3	Level 4	Level 5	Level 6	Level 7	Level 8	
Ideas and opportunities	Spotting opportunities	Use your imagination and abilities to identify opportunities for creating value.	Identify and seize opportunities to create value by exploring the social, cultural and economic landscape. Identify needs and challenges that need to be met. Establish new connections and bring together scattered elements of the landscape to create opportunities to create value.	Identify, create and seize opportunities.	I can find opportunities to help others.	I can recognise opportunities to create value in my community and surroundings.	I can explain what makes an opportunity to create value.	I can proactively look for opportunities to create value, including out of necessity.	I can describe different analytical approaches to identify entrepreneurial opportunities.	I can use my knowledge and understanding of the context to make opportunities to create value.	I can judge opportunities for creating value and decide whether to follow these up at different levels of the system I am working in (for example, micro, meso or macro).	I can spot and quickly take advantage of an opportunity.	
				Focus on challenges.	I can find different examples of challenges that need solutions.	I can recognise challenges in my community and surroundings that I can contribute to solving.	I can identify opportunities to solve problems in alternative ways.	I can redefine the description of a challenge, so that alternative opportunities address it may become apparent.	I can take apart established practices and challenge mainstream thought to create opportunities and look at challenges in different ways.	I can judge the right time to take an opportunity to create value.	I can cluster different opportunities or identify synergies among different opportunities to make the most out of them.	I can define opportunities where I can maintain a competitive advantage.	
				Uncover needs.	I can find examples of groups who have benefited from a solution to a given problem.	I can identify needs in my community and surroundings that have not been met.	I can explain that different groups may have different needs.	I can establish which user group, and which needs, I want to tackle through creating value.	I can carry out a needs analysis involving relevant stakeholders.	I can identify challenges related to the contrasting needs and interests of different stakeholders.	I can produce a "roadmap" which matches the needs with the actions needed to deal with them and helps me create value.	I can design projects which aim to anticipate future needs.	

2.4 How should limited professional development time be allocated?

Building teachers' own competences vs development of ability to teach them?

These are not alternatives.

Both are equally important and mutually supportive.

- Integrated learning outcomes approach to the development of competences is the way to go.
- Teach as we preach: e.g. teaching how to apply formative assessment in the classroom requires TPD that makes teachers experience formative assessment and analysing their own experience, developing “assessment as learning” mindset, practicing peer-learning and mastering a variety of techniques...



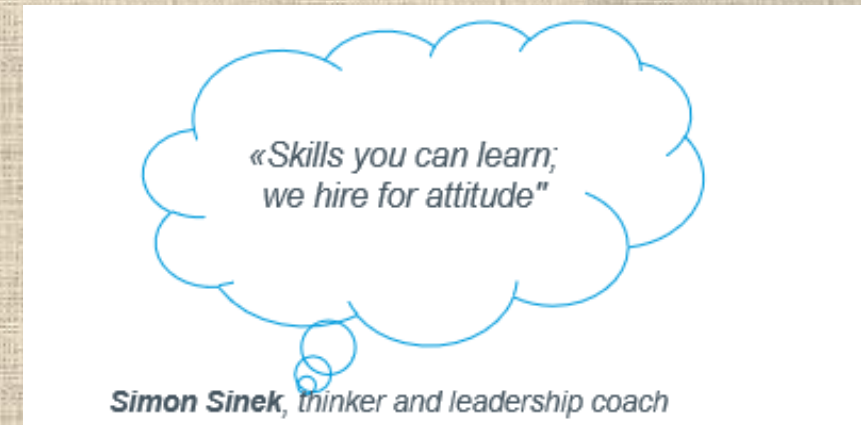
2.5 What is the role of teacher vulnerability and “not-knowing”?

Uncertainty and learning
vs. confident authority...

- Not knowing is a precious driver of progress in learning and its realization is growing as our competence increases...
- Up to a certain degree 😊 (as ignorant people cannot teach).
- Key learning outcomes comprising K/S/A are defined in the standards, while human knowledge could not be absolute these days.

2.6 How would we assess whether a teacher has “enough” competency?

- Flexible profiles. Flexible TPD models (e.g. micro-credentials). Link to teacher professional standards.
- Adequate performance evaluation approaches tuned to the learning outcomes for the priority teacher competences. Various competences are assessed differently. Formative vs. summative.



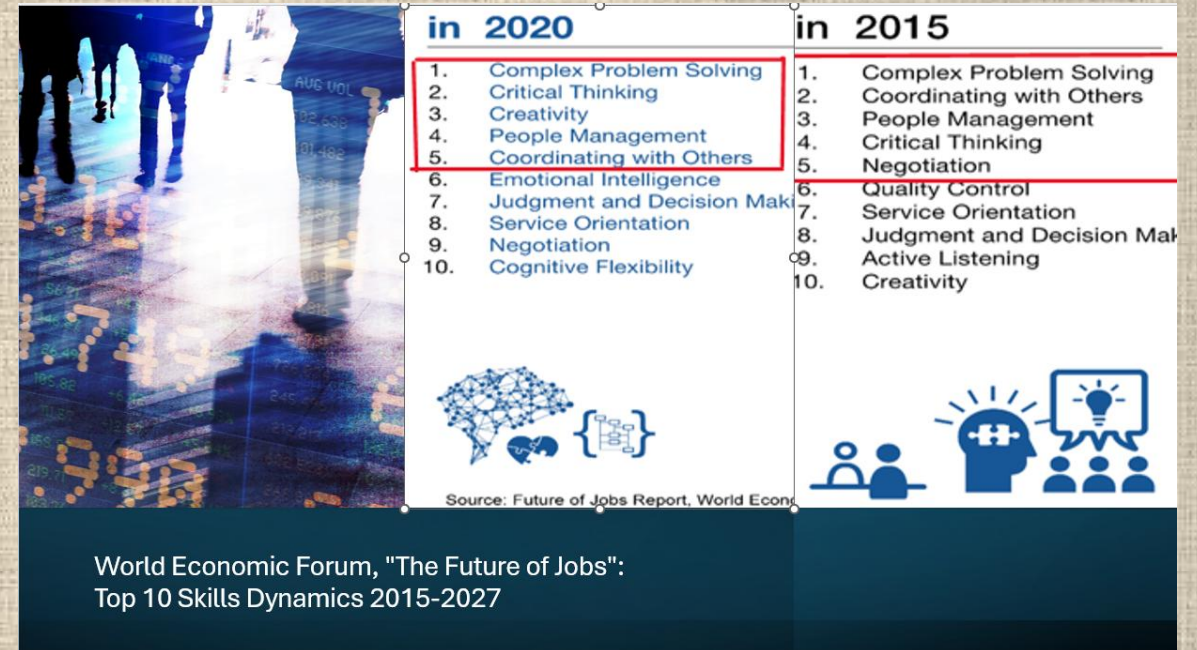
2.7 Does the answer change across contexts?

- Primary school vs. university
- STEM vs. humanities
- Across cultures and resource levels

Context is important and may impact learning objectives, learning strategies, techniques and tools...

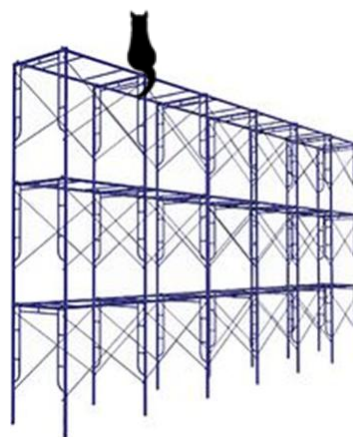
The lines between formal education and nonformal learning are blurring.

Teachers are lifelong learners. Professionals become teachers. Learners are drivers of their own learning in the learner-centered and learner-driven setting. It is important who we are.



Top skills for today and tomorrow:

Analytical thinking. Creative thinking.
Resilience, flexibility and agility.
Motivation. Leadership. Empathy.
Artificial intelligence and big data.



SCAFFOLD CARD DECK

**jointly with European Commission's Joint Research Centre (JRC)*



Scaffold gives easy access to competence frameworks



It helps visualise and mix competences



It helps design the teaching activity



It gives hints and ideas to teachers



Gamification of lesson planning can be motivational and enjoyable

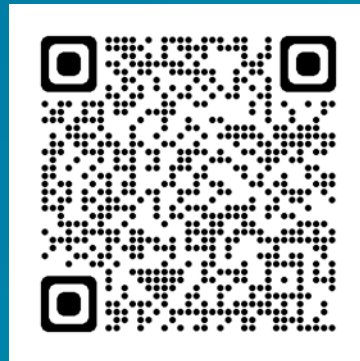
WHAT IS “SCAFFOLD” CARD DECK?

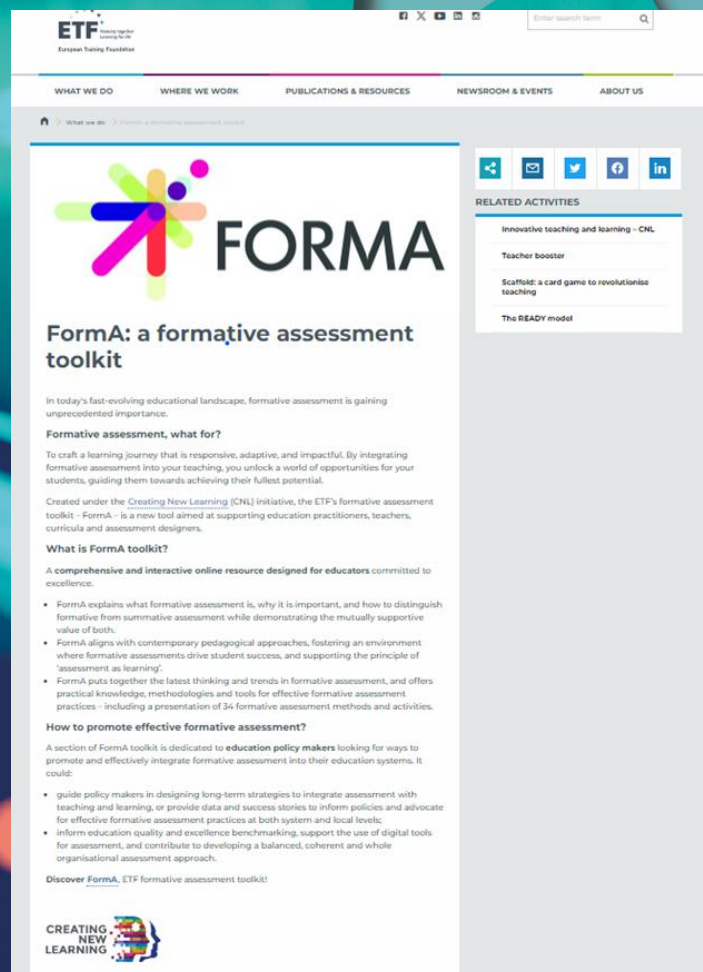
A deck of 102 cards for educators and a short User Guide

Applicable to both formal and non-formal learning, any subject area, any form of learning.

Helps teachers in designing any types of learning experience - from one lesson to a longer course.

Integrates and combines multiple key competences into the learning process.





EXPLORE THE TOOLKIT ON THE ETF WEBSITE:



<https://www.etf.europa.eu/en/what-we-do/forma-formative-assessment-toolkit>

**CREATING
NEW
LEARNING**



THANK YOU

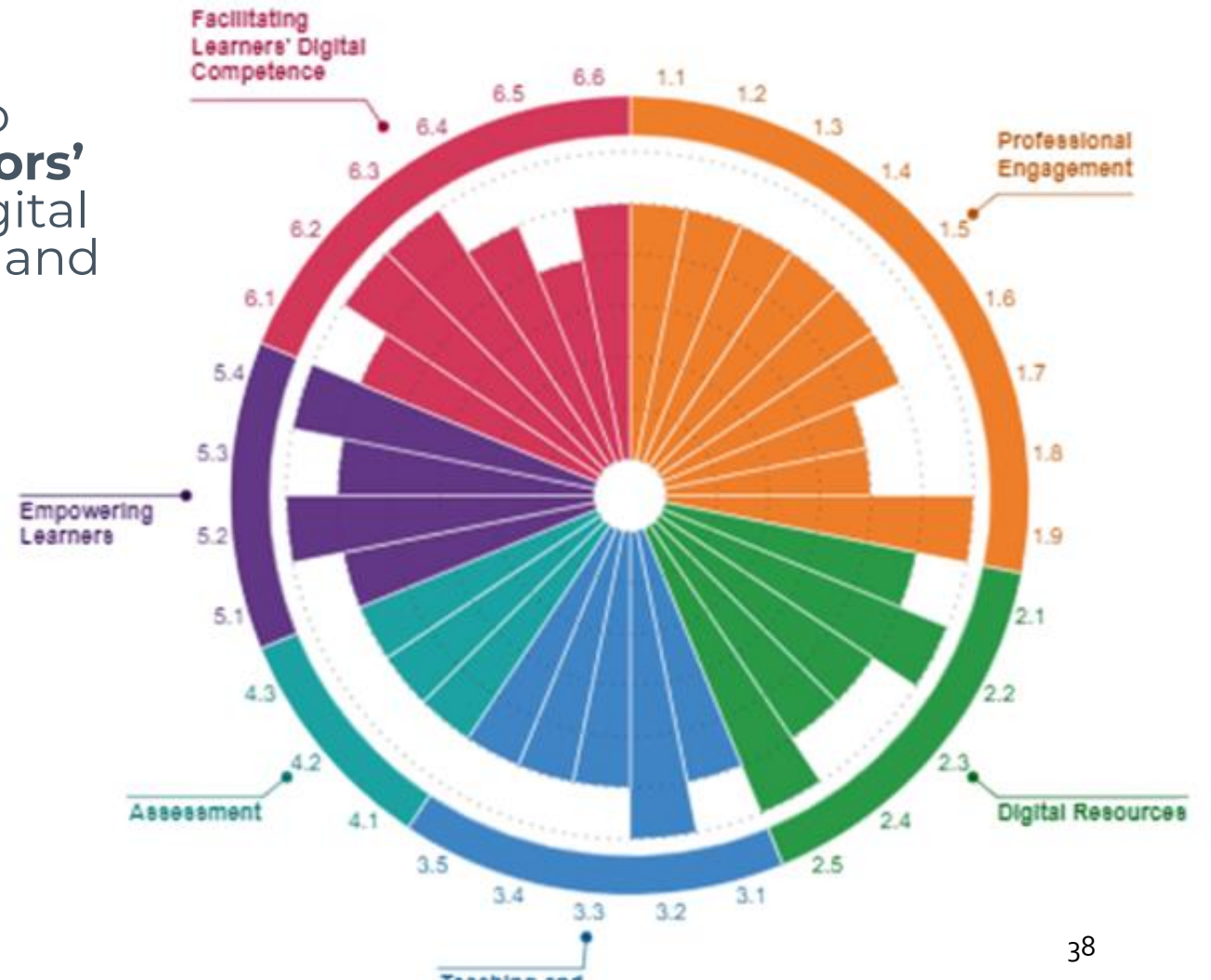
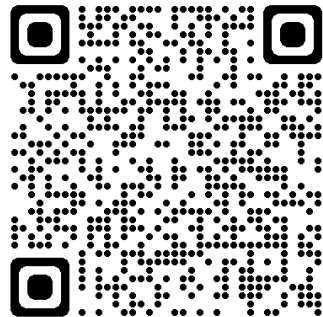
Email: Olena.Bekh@etf.europa.eu

Additional slides in case of questions

SELFIE FOR TEACHERS

- A free, online, self-reflection tool to support the **development of educators' digital competence** for the use of digital technology in teaching & assessment and for empowering learners

Watch the Video



SCAFFOLD WAS CREATED JOINTLY BY THE ETF AND JRC



Created jointly by the **European Training Foundation** (ETF) (under the Creating New Learning/CNL) and **Joint Research Centre** of the European Commission (JRC).

- Based on **FOUR European competence frameworks** for lifelong learning.
- Links together **digital, entrepreneurial, personal, social, and learning to learn competences**, and **competences for sustainability**, and six **transversal competences**.
- Used for reference and is a “**learning-by-doing**” tool for teachers.



Educators are free to select the cards they need and can build complexity as they master the tool.

SCAFFOLD IS DIVIDED INTO 5 GROUPS OF CARDS:

- 1** **Setting cards** (8 cards) to conceptualise the lesson design, the objective of learning, the learners' needs, duration, resources, etc.
- 2** **Planning cards** (7 cards) to guides the teacher through lesson design steps to choose one of the competences, teaching and assessment methods, set the timeline of the learning activity.
- 3** **Competence cards** (57 cards) representing the “library of key competences in the four EU frameworks: 21 for DigComp, 15 for EntreComp, 9 for LifeComp and 12 for GreenComp) plus **Transversal Competence cards** (7 cards) representing the competences that cut across the four EU frameworks.
- 4** **Teaching methods** (7 cards) giving examples of modern pedagogical approaches.
- 5** **Assessment methods** (7 cards) giving examples of various assessment methods.



Place them on a flat surface, or on a wall! You could work alone or in a group.

What is in the FormA Toolkit?

- **What is** formative assessment and why is it important?
- Differentiating **formative and summative** assessment
- The **importance** of formative assessment to learning
- The **ETF Formative Assessment Model**
- Formative assessment **methods and activities**
- What do **learners need** from formative assessment?
- What **can policy makers do** to promote and support effective formative assessment?

And also:

- **Latest thinking** and **trends** in formative assessment
- **Additional resources** and further reading

The ETF Model for Formative Assessment

Element 6

Using feedback to adapt teaching and learning, addressing gaps and learning needs.

6

5

Element 5

Delivering and receiving effective feedback

4

Element 4

Evaluating progress through diverse learning and assessment activities and evidence.

3

Element 3

Creating shared learning/problem-solving goals and success criteria.

2

Element 2

Empowering learners as assessors.

1

Element 1

Developing a whole organisational ethos and culture to encourage interaction with and reflection upon wide-ranging evidence of learning.

34 Formative assessment methods and activities

Peer observation

Ikigai as self-assessment

Comparative ranking

Setting and Sharing Learning and Developmental Goals

Co-Created Success Criteria for Task Clarification

Feynman Technique for Formative Assessment

Background Knowledge Probe for Continuous Feedback

Scrum

Structured questioning

Feedback Carousel

Peer Quizzes

Incidental Scenarios for Mindset Differentiation

Using Journals for learning reflection

Traffic Light Feedback

2 stars and a goal

SMART audit

Formative assessment of video evidence

Pitching Progress in 30 Seconds

Exit Poll

Implementing Learner-Centred Exhibitions for authentic feedback

Together We Learn More

How would you? Applying formative feedback to future scenarios

Tripwires

Affirmation Cards for Growth Mindset

5 Whys and Hows

3-2-1 Activity

Truths and a Lie for Formative Assessment

Even better if...

Theory of Change for Formative Assessment

Ipsative Assessment

Spot the difference

Peer mentoring

Formative Feedback loop workshop

The power of yet

What can Policy makers do to promote effective formative assessment?

- ✓ Long-Term Strategy for Assessment Integration
- ✓ Using digital tools for formative assessment
- ✓ Developing a Balanced, Coherent and Whole Organisational Assessment Approach
- ✓ Focusing on 'Fit for Purpose' Assessment
- ✓ Integrating Assessment with Teaching and Learning